

DIGITAL FORENSICS

Correlating Event Logs with Registry Hives Using GKAPE and Eric Zimmerman Tools

Case Reference No.	DFIR-2026-ELRC-011
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Date of Analysis	August 26, 2026
Prepared For	Public
Platform	Windows
Tools	GKAPE, EvtxECmd, RECcmd

Table of Contents

Case Information	Error! Bookmark not defined.
Table of Contents.....	2
1. Objectives	3
2. Table of Evidence Sources	3
3. Phase 2: Verify Evidence Collection	5
4. Phase 3: Parse the Event Logs and Registry Hives.....	8
Step 1: Parse Event Logs	8
Step 2: Parse the Registry Hive	8
5. Normal Startup vs Unexpected Shutdown.....	8
Expected Normal Sequence	8
What I Observed.....	8
6. Correlating Event Viewer with Parsed Results.....	9
7. Findings	11
8. Post-Examination Integrity Verification.....	11
9. Conclusion	12

1. Objectives

The purpose of this exercise is for me to correlate Windows Event Logs with Registry Hive artifacts to reconstruct system activity. My objectives are to:

- Establish timelines of system startup, shutdown, and restart events.
- Correlate user activity with recorded system events.
- Identify abnormal or unexpected shutdowns.
- Verify whether system reboots were user initiated.

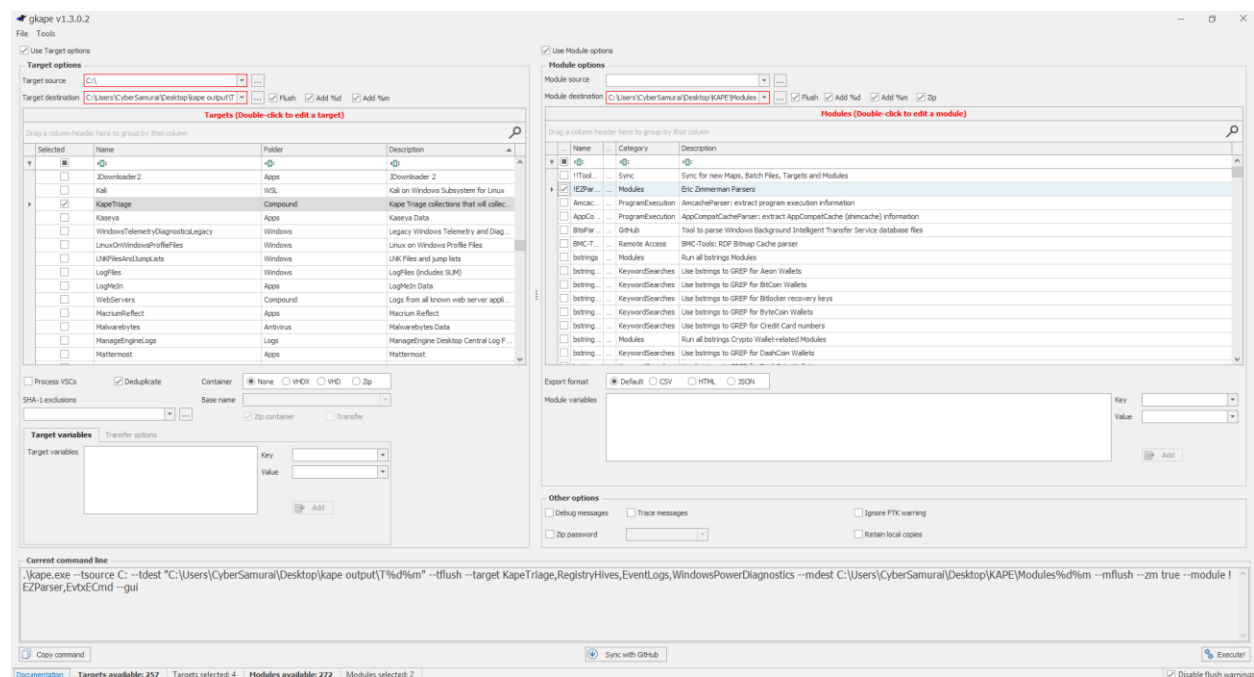
2. Table of Evidence Sources

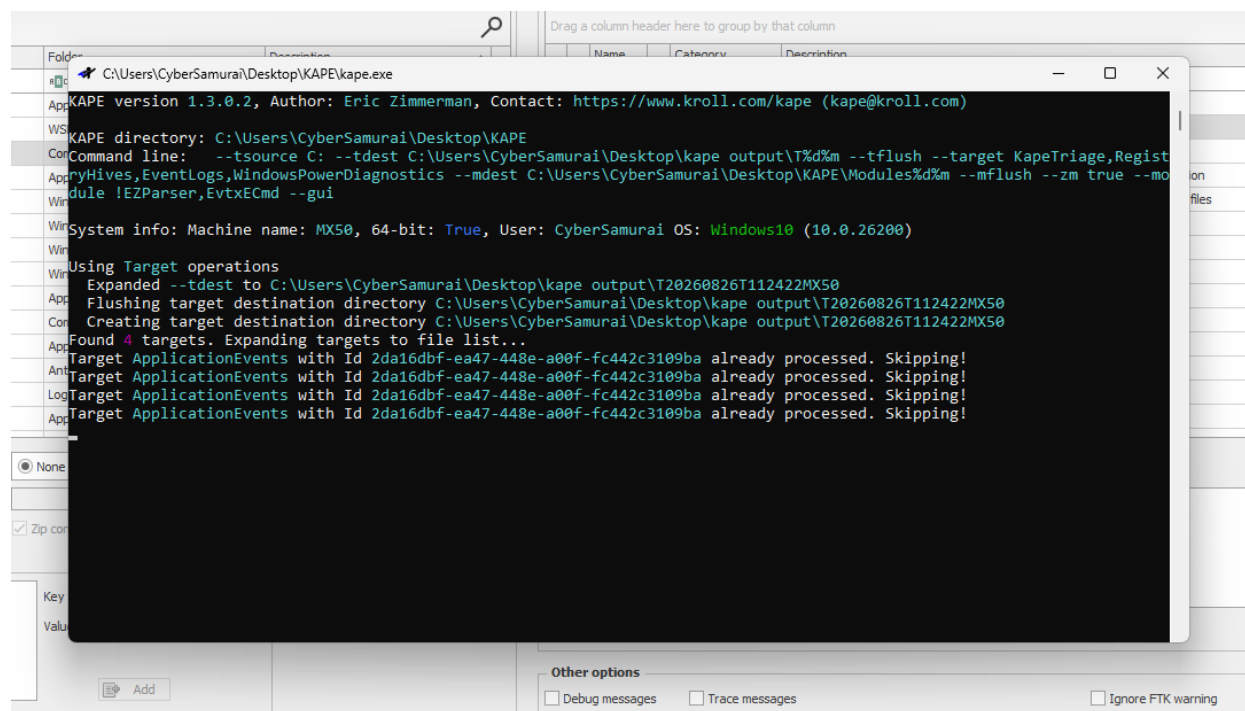
Evidence Source	Acquisition Tool	Hashing Tool
C:	GKAPE	PowerShell

I ran GKAPE as Administrator and selected my source to be C:\.

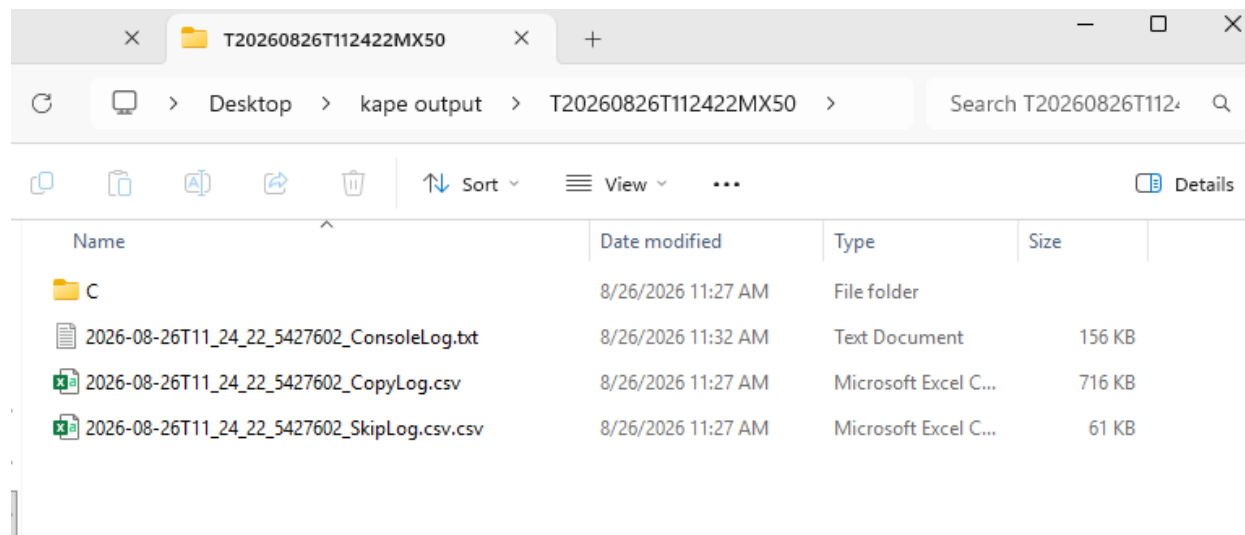
Next, I selected my destination folders for both Targets and Modules.

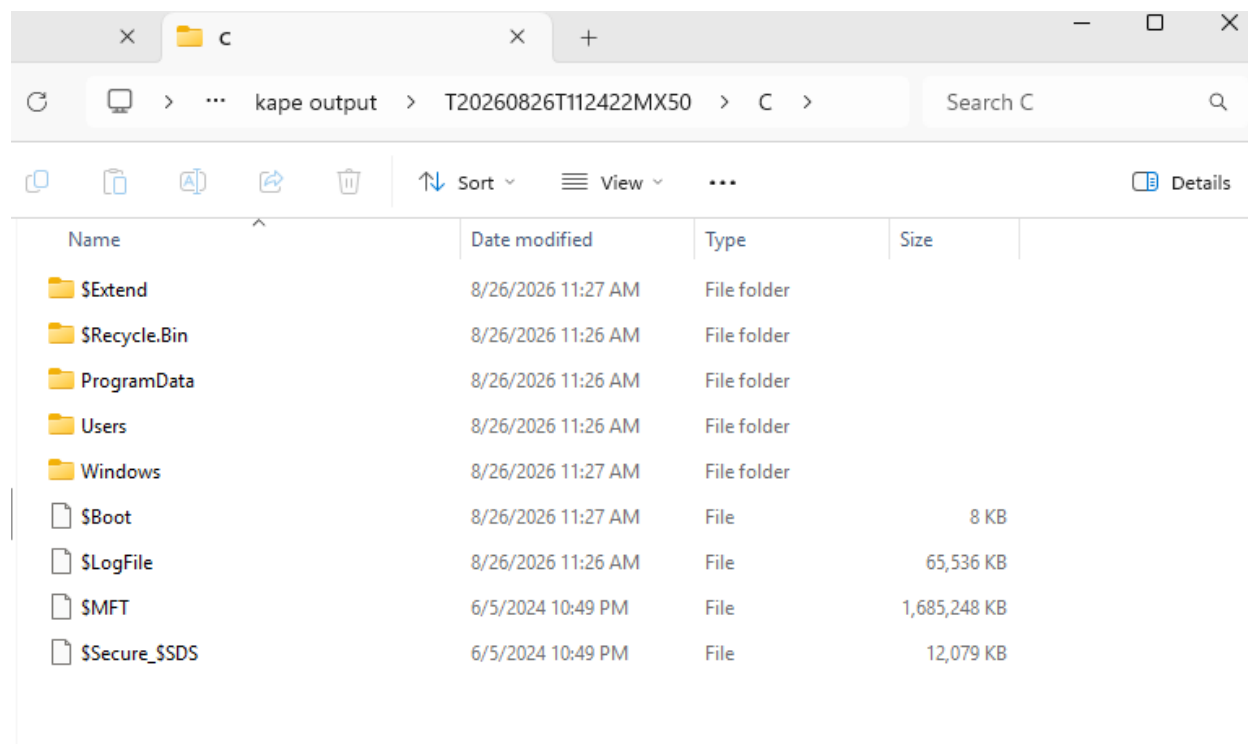
Targets	Modules
KapeTriage	!EZParser
Windows Event Logs	



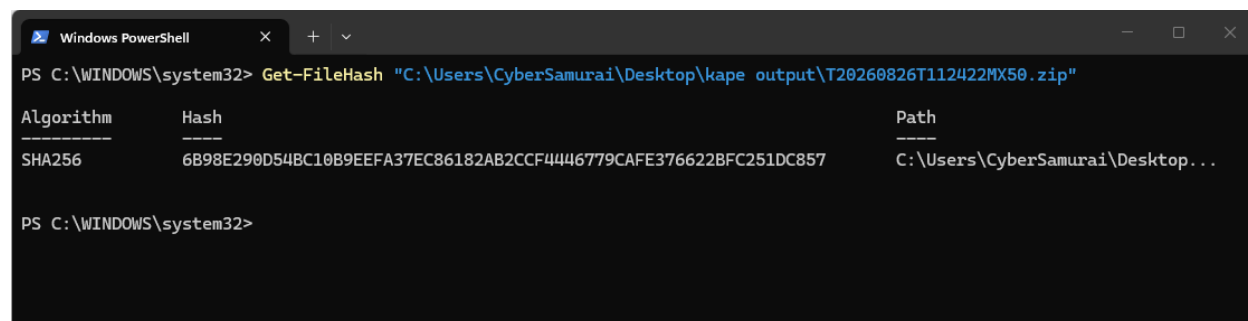


GKAPE Output Folder





GKAPE Output Folder Hashed using PowerShell



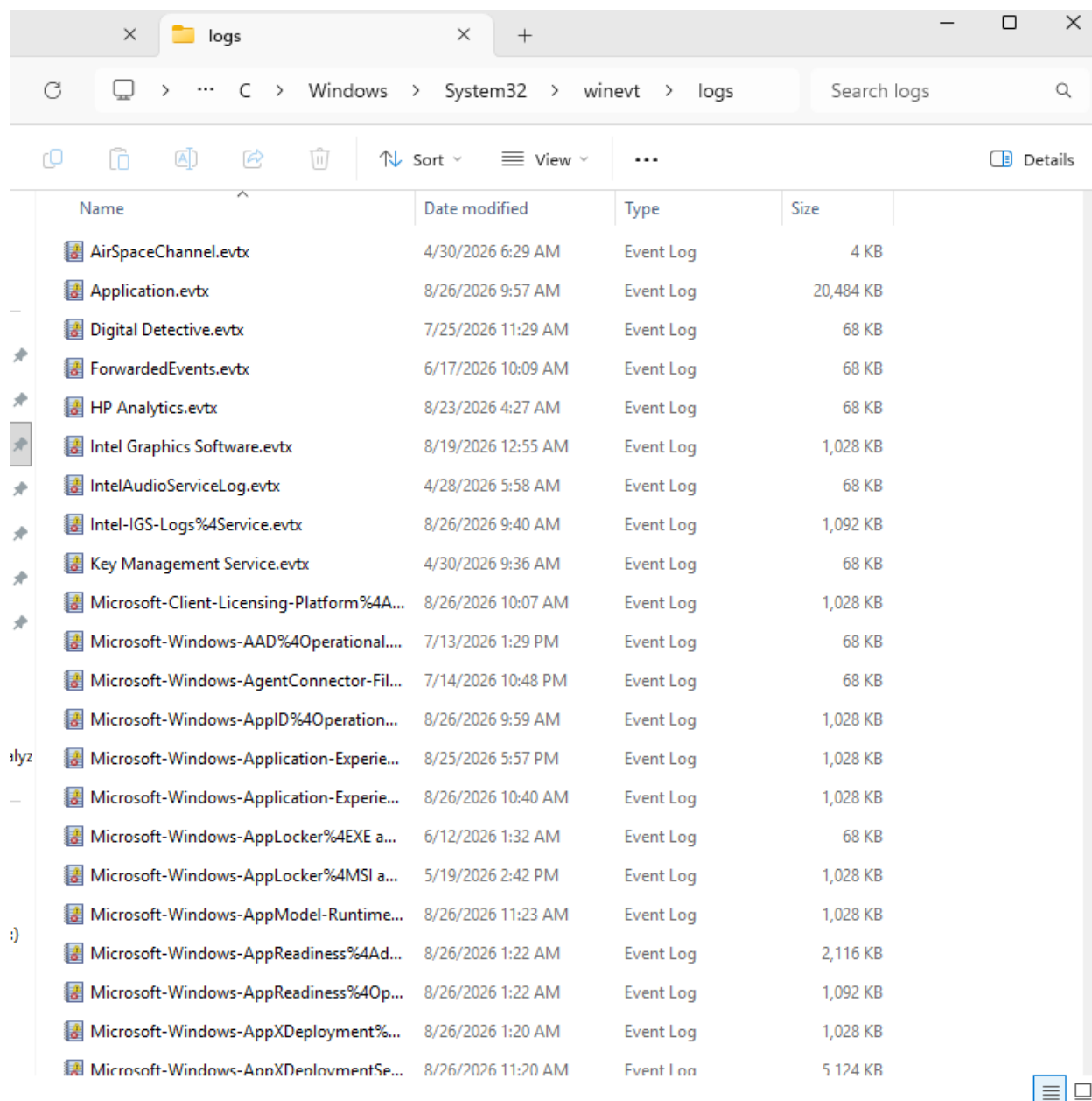
3. Phase 2: Verify Evidence Collection

I confirmed that the required evidence was successfully acquired in the following directories:

1. C:\Windows\System32\winevt\Logs to verify Event Logs.
2. C:\Windows\System32\config to verify Registry Hives.
3. C:\Users\CyberSamurai\NTUSER.DAT to verify the user registry hive.

Evidence Verification

Event Logs folder



Name	Date modified	Type	Size
AirSpaceChannel.evtx	4/30/2026 6:29 AM	Event Log	4 KB
Application.evtx	8/26/2026 9:57 AM	Event Log	20,484 KB
Digital Detective.evtx	7/25/2026 11:29 AM	Event Log	68 KB
ForwardedEvents.evtx	6/17/2026 10:09 AM	Event Log	68 KB
HP Analytics.evtx	8/23/2026 4:27 AM	Event Log	68 KB
Intel Graphics Software.evtx	8/19/2026 12:55 AM	Event Log	1,028 KB
IntelAudioServiceLog.evtx	4/28/2026 5:58 AM	Event Log	68 KB
Intel-IGS-Logs%4Service.evtx	8/26/2026 9:40 AM	Event Log	1,092 KB
Key Management Service.evtx	4/30/2026 9:36 AM	Event Log	68 KB
Microsoft-Client-Licensing-Platform%4A...	8/26/2026 10:07 AM	Event Log	1,028 KB
Microsoft-Windows-AAD%4Operational....	7/13/2026 1:29 PM	Event Log	68 KB
Microsoft-Windows-AgentConnector-Fil...	7/14/2026 10:48 PM	Event Log	68 KB
Microsoft-Windows-AppID%4Operation...	8/26/2026 9:59 AM	Event Log	1,028 KB
Microsoft-Windows-Application-Experie...	8/25/2026 5:57 PM	Event Log	1,028 KB
Microsoft-Windows-Application-Experie...	8/26/2026 10:40 AM	Event Log	1,028 KB
Microsoft-Windows-AppLocker%4EXE a...	6/12/2026 1:32 AM	Event Log	68 KB
Microsoft-Windows-AppLocker%4MSI a...	5/19/2026 2:42 PM	Event Log	1,028 KB
Microsoft-Windows-AppModel-Runtime...	8/26/2026 11:23 AM	Event Log	1,028 KB
Microsoft-Windows-AppReadiness%4Ad...	8/26/2026 1:22 AM	Event Log	2,116 KB
Microsoft-Windows-AppReadiness%4Op...	8/26/2026 1:22 AM	Event Log	1,092 KB
Microsoft-Windows-AppXDeployment%...	8/26/2026 1:20 AM	Event Log	1,028 KB
Microsoft-Windows-AnnXDeploymentSe...	8/26/2026 11:20 AM	Event Log	5,124 KB

Registry Hives (SYSTEM, SOFTWARE, SAM, SECURITY)

config

C > Windows > System32 > config

Search config

Name	Date modified	Type	Size
DEFAULT	8/20/2026 11:01 PM	File	1,280 KB
DEFAULT.LOG1	4/28/2026 6:39 AM	LOG1 File	400 KB
DEFAULT.LOG2	4/28/2026 6:39 AM	LOG2 File	355 KB
SAM	8/20/2026 11:01 PM	File	128 KB
SAM.LOG1	4/28/2026 6:39 AM	LOG1 File	64 KB
SAM.LOG2	4/28/2026 6:39 AM	LOG2 File	64 KB
SECURITY	8/20/2026 11:01 PM	File	64 KB
SOFTWARE	8/21/2026 10:09 AM	File	147,968 KB
SOFTWARE.LOG1	4/28/2026 6:39 AM	LOG1 File	19,456 KB
SOFTWARE.LOG2	4/28/2026 6:39 AM	LOG2 File	23,296 KB
SYSTEM	8/20/2026 11:01 PM	File	66,816 KB
SYSTEM.LOG1	4/28/2026 6:39 AM	LOG1 File	4,160 KB
SYSTEM.LOG2	4/28/2026 6:39 AM	LOG2 File	16,174 KB

NTUSER.DAT confirmation

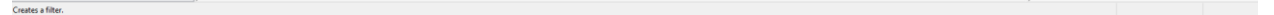
CyberSamurai

C > Users > CyberSamurai

Search CyberSamurai

Name	Date modified	Type	Size
AppData	8/26/2026 11:26 AM	File folder	
NTUSER.DAT	8/24/2026 9:46 AM	DAT File	11,776 KB
ntuser.dat.LOG1	4/27/2026 11:08 PM	LOG1 File	4,152 KB
ntuser.dat.LOG2	4/27/2026 11:08 PM	LOG2 File	2,945 KB

Event ID	Date & Time	Interpretation
12	6/26/2026 10:49:32 AM	Operating system started
6005	6/26/2026 10:49:32 AM	Event Log service started
41	6/26/2026 10:49:37 AM	Unexpected power loss
6008	6/26/2026 10:49:58 AM	Unexpected shutdown



Filtered Event Viewer results showing Event IDs 12, 41, 6005 and 6008

7. Findings

Objective 1: Establish Timelines

Status: Achieved

I used the combination of Event IDs 12, 6005, 41, 6008, and 1074 to establish the sequence of startup, shutdown, and reboot activity.

Objective 2: Prove User Activity

Status: Partially Achieved

I found that Event ID 1074 records that the Windows account MX50\CyberSamurai initiated a restart on 3 Jun 2026 at 10:51:41 PM. This associates a user account with the reboot event.

Objective 3: Detect Abnormal Shutdowns

Status: Achieved

I determined that Event ID 41 followed by 6008, together with the absence of 13 and 6006, demonstrates that Windows experienced an unexpected shutdown.

Objective 4: Verify Reboots

Status: Achieved

I confirmed that Event ID 1074 shows a graceful, user-initiated restart and provides the exact timestamp of the reboot.

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8. Post-Examination Integrity Verification

```
Windows PowerShell
PS C:\WINDOWS\system32> Get-FileHash "C:\Users\CyberSamurai\Desktop\kape_output\T20260826T112422MX50.zip"

Algorithm      Hash
-----
SHA256         6B98E290D54BC10B9EEFA37EC86182AB2CCF4446779CAFE376622BFC251DC857
Path           C:\Users\CyberSamurai\Desktop\...

PS C:\WINDOWS\system32> Get-FileHash "C:\Users\CyberSamurai\Desktop\kape_output\T20260826T112422MX50.zip"

Algorithm      Hash
-----
SHA256         6B98E290D54BC10B9EEFA37EC86182AB2CCF4446779CAFE376622BFC251DC857
Path           C:\Users\CyberSamurai\Desktop\...

PS C:\WINDOWS\system32>
```

9. Conclusion

This examination demonstrated the collection, verification, parsing, and correlation of Windows Event Logs and Registry Hives using GKAPE, EvtxECmd, and RECcmd.

GKAPE was used to acquire the relevant Windows forensic artifacts. EvtxECmd was used to parse the Windows Event Logs, while RECcmd was used to examine the Registry Hives.

The Event Log analysis established evidence of system startup, unexpected shutdown activity, and a user-associated restart. Event IDs 41 and 6008 indicated an unexpected shutdown, while Event ID 1074 identified MX50\CyberSamurai as the account associated with a restart on 3 Jun 2026 at 10:51:41 PM.

The original Event Viewer records were also compared with the parsed EvtxECmd output to validate the identified Event IDs and timestamps.

The correlation of Event Logs and Registry evidence provides a stronger forensic reconstruction than relying on either evidence source independently. However, conclusions regarding malicious tampering or physical user presence require additional supporting evidence and should not be inferred from the Event IDs alone.